

24m0847 (Vansil Rakeshkumar Chauhan)

Part 5a:

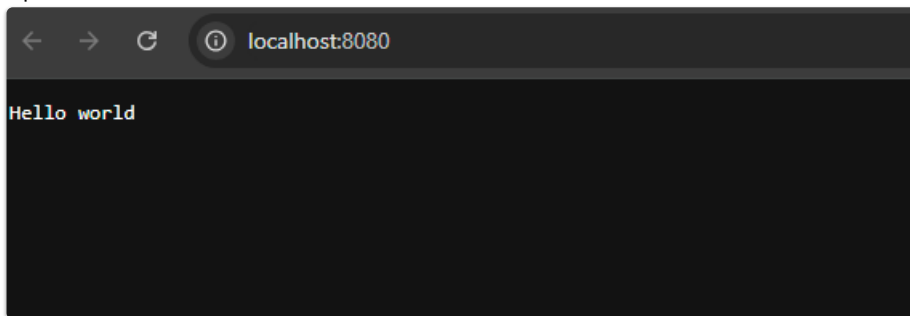
Instructions to execute:

1. `cflask` server with 10 threads and on port number 8080 :

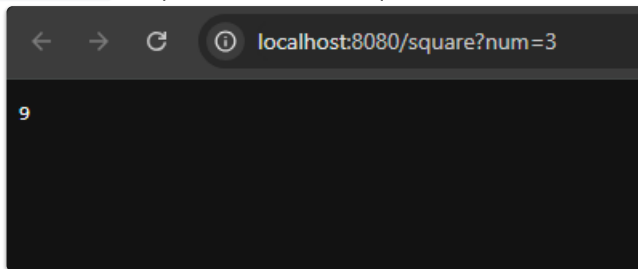
```
# make clean; make; ./cflask <port> <thread_count>
make clean; make; ./cflask 8080 10
```

2. Client (examples) (click link):

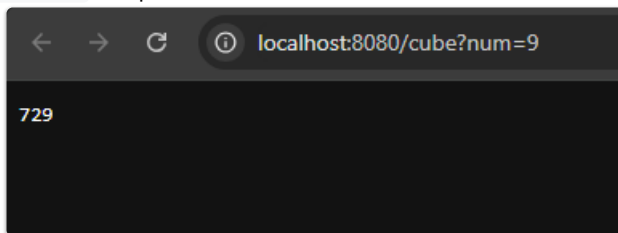
1. `/` : <http://localhost:8080/>



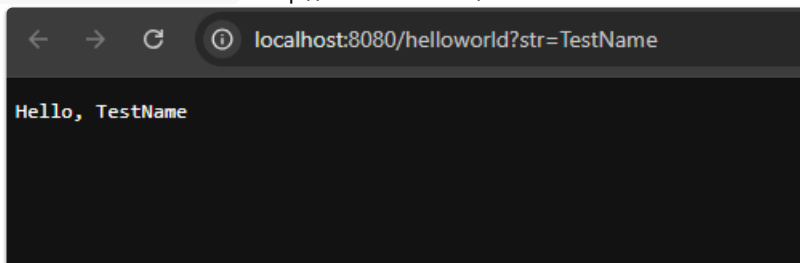
2. `/square?num=3` : <http://localhost:8080/square?num=3>



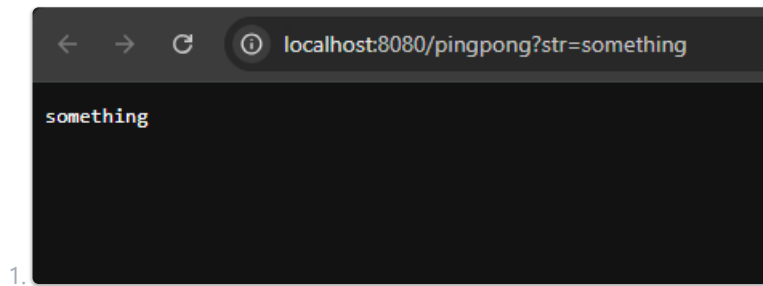
3. `/cube?num=9` : <http://localhost:8080/cube?num=9>



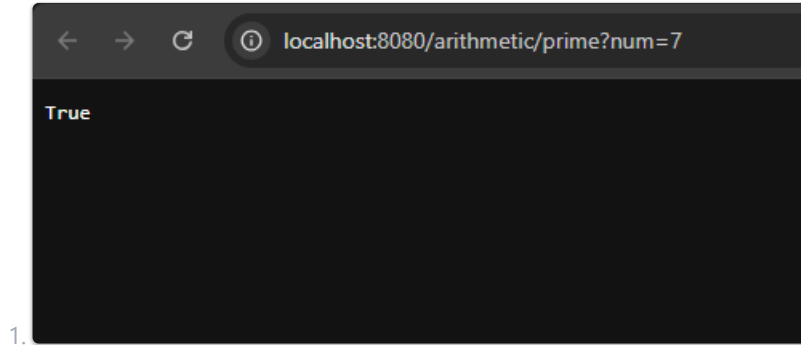
4. `/helloworld?str=TestName` : <http://localhost:8080/helloworld?str=TestName>



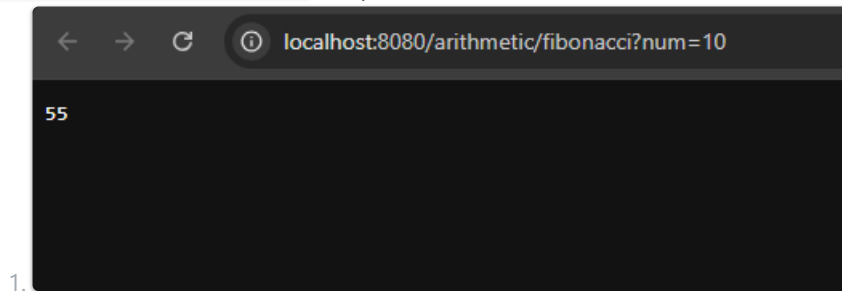
5. `/pingpong?str=something` : <http://localhost:8080/pingpong?str=something>



6. `/arithmetic/prime?num=7` : `http://localhost:8080/arithmetic/prime?num=7`



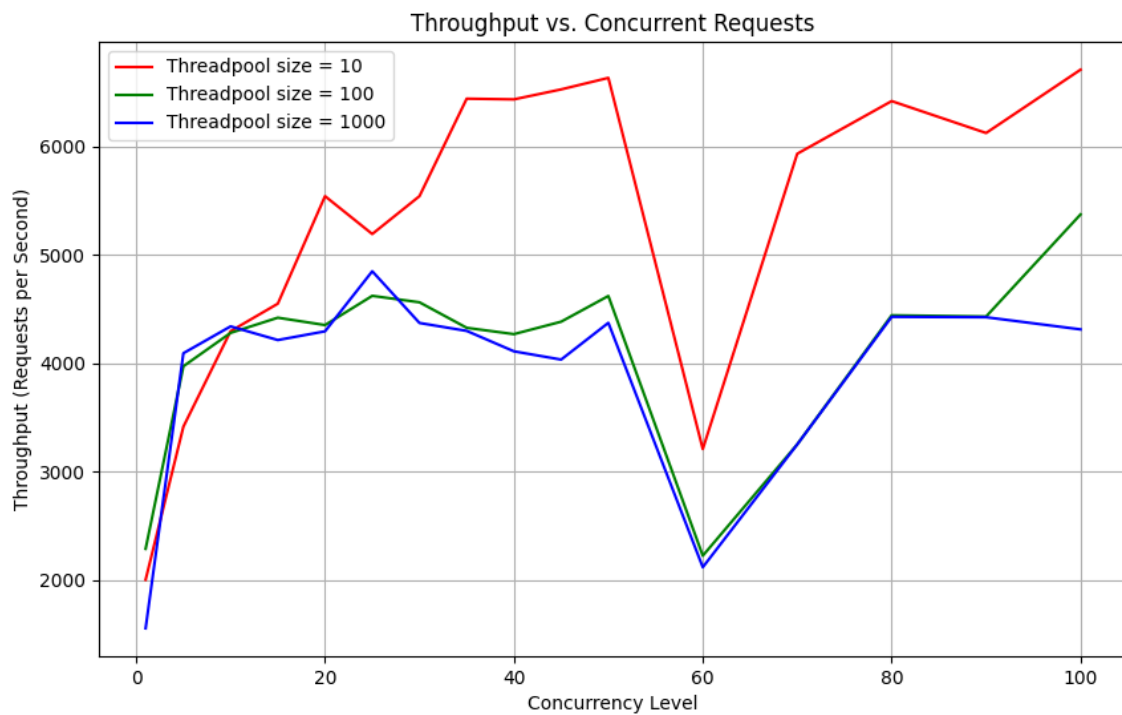
7. `/arithmetic/fibonacci?num=10` : `http://localhost:8080/arithmetic/fibonacci?num=10`



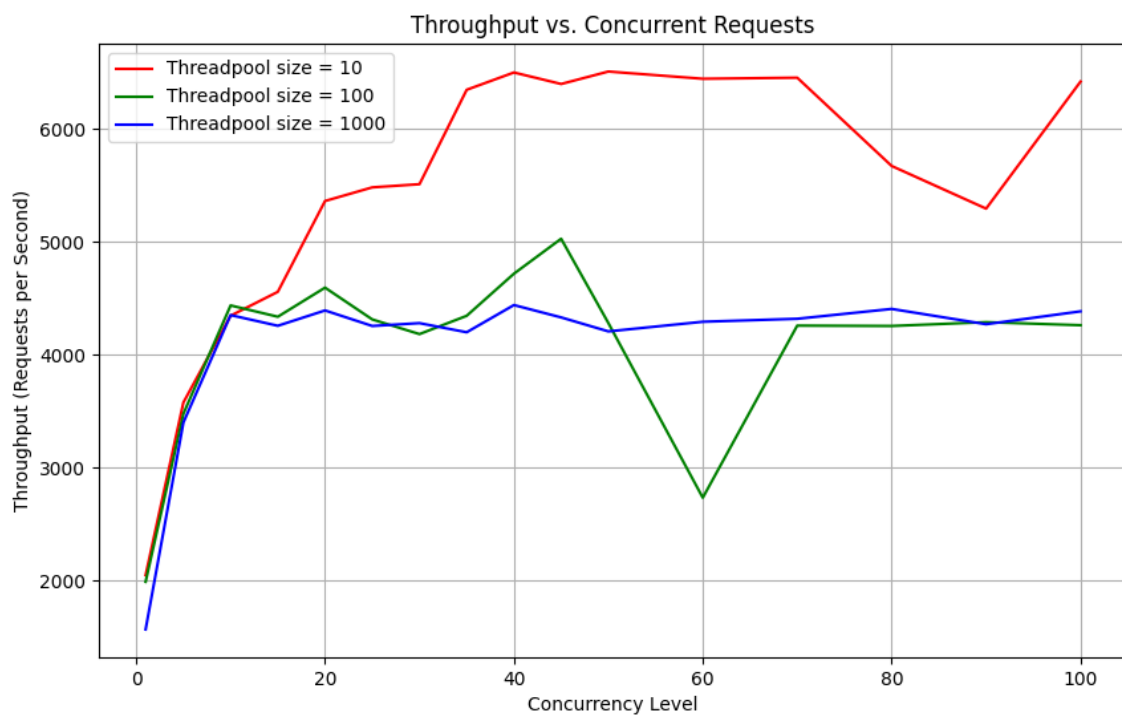
Part 5b:

Graphs:

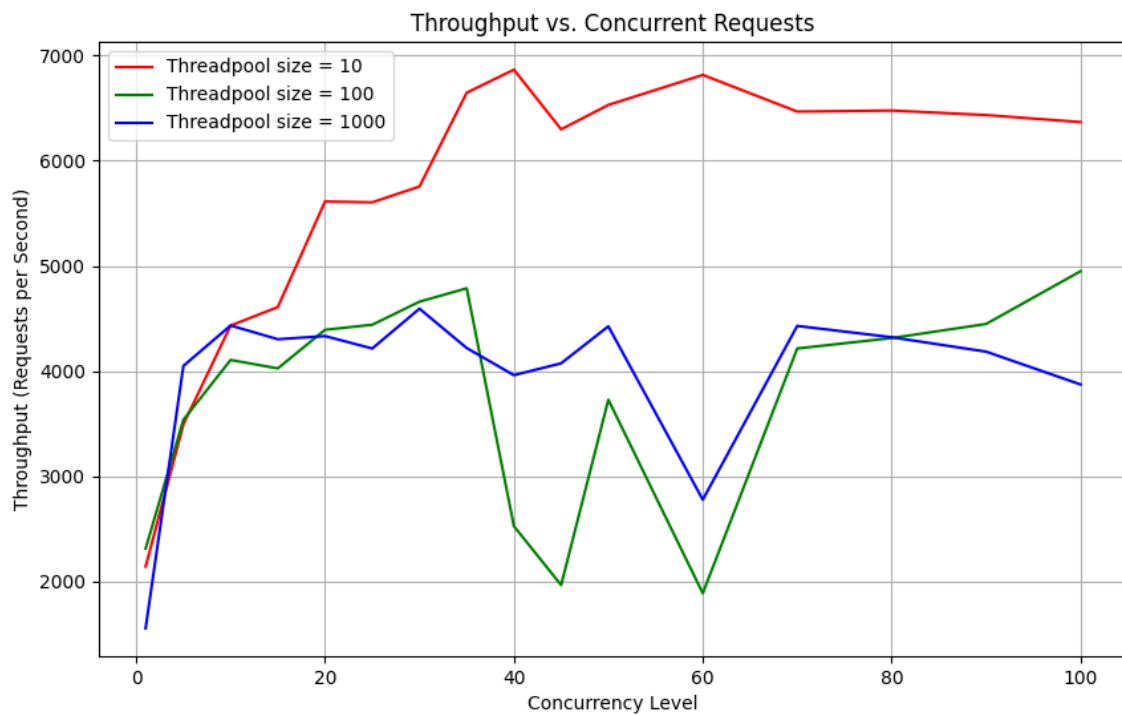
`http://localhost:8080/arithmetic/prime?num=2001`



<http://localhost:8080/arithmetic/fibonacci?num=35>



<http://localhost:8080/helloworld?str=Vansil:>



Reporting of Experiments:

A: aim/purpose of the experiment

1. The purpose of this experiment is to evaluate how the server handles varying load conditions by analyzing performance metrics under different scenarios. We aim to understand the server's behavior in response to concurrent requests and observe any limitations in its handling capacity.

B: setup and execution details, metrics and independent parameter

1. In this experiment, we used a load-testing tool to generate varying levels of load on the server. We set up the environment with defined configurations and executed tests across multiple load levels, adjusting a specific parameter to observe changes in server behavior. Metrics such as responsiveness, throughput, and reliability were tracked, while the main variable was adjusted to see its impact on performance.

C: hypothesis/expectation

1. We anticipate that server performance will fluctuate based on the changes to the primary variable, with certain configurations likely yielding optimal results. As the load increases, we expect some performance degradation, with specific thresholds leading to observable bottlenecks or slowdowns.

D: observations from the data/plots

1. From the collected data and generated plots, we observed how server performance metrics responded to different levels of load. Patterns emerged showing how various configurations affected key performance indicators, and certain trends appeared that were aligned or misaligned with our expectations.

E: explanation of behaviour and inferences

1. Analyzing the data revealed insights into the server's capacity and efficiency under load. Specific configurations led to expected behaviors, while others exposed limitations or inefficiencies. These findings helped identify possible areas for improvement and strategies to optimize server performance based on the observed behaviors.